Heinrich Weber, *Lehrbuch der Algebra*, Volume II

English Translation: Sections 155–175

Source pp. 559–640

Eighteenth Section. Discriminants

§155. The Minimum of a Quadratic Form

The finiteness theorem for the classes of ideals, which was obtained earlier by a purely arithmetical argument, may be reached in another and very fruitful way. The method, due to Minkowski, depends first on a theorem about positive quadratic forms. We state and prove it in the form in which it is needed for algebraic number fields.

Let

$$f(x_1, \ldots, x_n) = \sum_{i,k} a_{ik} x_i x_k$$

be a real quadratic form which is positive for every non-zero system of real values x_i. It is then possible to write it as a sum of n squares of real linear forms,

$$f = \xi_1^2 + \xi_2^2 + \cdots + \xi_n^2, \qquad \xi_i = \alpha_{1i} x_1 + \alpha_{2i} x_2 + \cdots + \alpha_{ni} x_n.$$

If A denotes the determinant of the coefficients α_{ij}, and if D denotes the determinant of the quadratic form f, then

$$D = A^2.$$

The inverse linear substitution has the form

$$x_i = \beta_{1i}\xi_1 + \beta_{2i}\xi_2 + \cdots + \beta_{ni}\xi_n \qquad (i = 1, \ldots, n).$$

When the x_i range over all rational integral systems, the corresponding points

$$(\xi_1, \ldots, \xi_n)$$

form a lattice in n-dimensional space. If the system $x_1, \ldots, x_n$ is not identically zero, then f assumes positive values; let M be the least of these values. Thus the least distance between two distinct lattice-points, measured in the ordinary Euclidean metric of the ξ-space, is $\sqrt{M}$.

Consider in the ξ-space an arbitrary bounded region G_ξ, and let its volume be

$$V_\xi = \int \cdots \int d\xi_1 \cdots d\xi_n.$$

Under the linear change of variables this volume is changed in the ratio $|A|$, so that

$$V_\xi = \sqrt{D}\, V_x.$$

In particular, if K_n is the volume of the unit sphere

$$\xi_1^2 + \cdots + \xi_n^2 \leq 1,$$

then a sphere of radius r has volume $K_n r^n$.

Around every lattice point draw a sphere of radius $\frac{1}{2}\sqrt{M}$. No two of these spheres have an interior point in common; for otherwise two lattice-points would have distance smaller than $\sqrt{M}$. Now take a large region similar to a fixed region G_ξ, and count the lattice-points in it. The quotient of the number of lattice-points by the volume tends, in the limit, to the reciprocal of the fundamental volume $\sqrt{D}$. Hence the spheres just constructed cannot, in the limiting sense, occupy more than the whole space. Therefore

$$\frac{K_n \left(\frac{1}{2}\sqrt{M}\right)^n}{\sqrt{D}} \leq 1,$$

and consequently

$$M \leq 4 \left(\frac{D}{K_n^2}\right)^{1/n}.$$

For the applications the strict inequality is sufficient:

$$M < 4 \left(\frac{D}{K_n^2}\right)^{1/n}.$$

A rough but simple estimate of K_n already gives a useful form of the theorem. The sphere of radius 1 contains the cube

$$|\xi_i| \leq \frac{1}{\sqrt{n}} \qquad (i = 1, \ldots, n),$$

whose volume is $(2/\sqrt{n})^n$. Hence

$$K_n > \left(\frac{2}{\sqrt{n}}\right)^n,$$

and from the preceding inequality follows

$$M < n\sqrt[n]{D}.$$

For $n > 1$ the inequality is strict:

$$M < n\sqrt[n]{D}. \tag{1}$$

A sharper estimate, which will be useful below, is obtained recursively. Cut the unit sphere by a cylinder with axis in the direction of the first coordinate. If $0 < a < 1$, the slice

$$|\xi_1| \leq a$$

contains, for each ξ_1 in that interval, an $(n-1)$-sphere of radius at least $\sqrt{1-a^2}$. Hence

$$K_n > 2a(1-a^2)^{(n-1)/2} K_{n-1}.$$

Choosing a so as to make the right hand side as large as possible gives a decreasing sequence of constants θ_n for which the preceding estimate may be written

$$M < \theta_n \, n \sqrt[n]{D}.$$

In dimension 2, since $K_2 = \pi$, one obtains

$$\theta_2 = \frac{2}{\pi} < 0.636, \qquad \theta_2^2 < 0.404.$$

Since the numbers θ_n decrease, we have, for every $n > 1$,

$$M < n \sqrt[n]{0.404\, D}. \tag{2}$$

Thus every positive quadratic form in n variables with determinant D assumes, for some non-zero integral system $x_1, \ldots, x_n$, a value smaller than the right hand side of (2). This is the form of Minkowski's theorem that will be applied to algebraic integers.

§156. Application to Algebraic Fields

We first record the elementary inequality that will be used in passing from sums of conjugate absolute values to norms. If $a_1, \ldots, a_n$ are positive real numbers, then

$$a_1 + \cdots + a_n \geq n(a_1 a_2 \cdots a_n)^{1/n}. \tag{1}$$

The proof is by induction. For $n = 2$ it is the identity

$$a + b - 2\sqrt{ab} = (\sqrt{a} - \sqrt{b})^2 \geq 0.$$

The general case follows by applying the assertion first to $n-1$ of the numbers and then to the two quantities so obtained.

Let Ω be an algebraic field of degree n, with minimal basis

$$\omega_1, \ldots, \omega_n$$

and field discriminant Δ. Let φ be a whole functional in the ring of integers o of Ω. Choose a basis of φ,

$$\beta_i = h_{1i}\omega_1 + h_{2i}\omega_2 + \cdots + h_{ni}\omega_n \qquad (i = 1, \ldots, n),$$

so that the determinant $h = (h_{ij})$ has absolute value equal to the absolute norm of the functional:

$$N_a(\varphi) = |\det(h_{ij})|.$$

Every whole number divisible by φ is then of the form

$$\eta = \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n,$$

where $x_1, \ldots, x_n$ are rational integers.

Let

$$\eta_1, \ldots, \eta_n$$

be the conjugates of η. We form the positive quadratic expression

$$f(x_1, \ldots, x_n) = |\eta_1|^2 + |\eta_2|^2 + \cdots + |\eta_n|^2. \tag{2}$$

For a real conjugate this is simply the square of a real linear form. For a pair of conjugate imaginary fields the product $\eta_i \eta_{i'}$ is the sum of two squares of real linear forms. Therefore (2) is a positive quadratic form in n real variables.

The determinant of this form is, apart from the usual sign of the field discriminant,

$$|D| = |\,\Delta\,|\, N_a(\varphi)^2.$$

Applying the theorem of the preceding section, we can choose the rational integers x_i, not all zero, so that

$$\sum_{i=1}^{n} |\eta_i|^2 < \theta_n\, n \sqrt[n]{|\Delta|\, N_a(\varphi)^2}, \tag{3}$$

where $\theta_n^n < 0.404$ for $n > 1$.

By (1),

$$n\, |N(\eta)|^{2/n} \leq |\eta_1|^2 + \cdots + |\eta_n|^2.$$

Combining this with (3) gives

$$|N(\eta)|^2 < \theta_n^n\, |\Delta|\, N_a(\varphi)^2. \tag{4}$$

Since η is divisible by φ, we may write, in the language of functionals,

$$\eta = \varphi\psi$$

up to a unit factor; consequently

$$|N(\eta)| = N_a(\varphi)\, N_a(\psi).$$

It follows from (4) that

$$N_a(\psi) < \sqrt{\theta_n^n |\Delta|} < \sqrt{0.404\,|\Delta|}. \tag{5}$$

Thus we obtain Minkowski's fundamental theorem:

For every ideal class in an algebraic field Ω of degree $n > 1$ there is an integral ideal belonging to that class whose absolute norm is smaller than

$$\sqrt{|\,\Delta\,|}. \tag{6}$$

Indeed, take φ in the inverse class. The construction gives ψ in the desired class, and (5) is stronger than (6).

A first consequence is the following. No algebraic field of degree $n > 1$ can have discriminant $\Delta = \pm 1$. For if $|\Delta| = 1$, the theorem would give, in every ideal class and in particular in the principal class, an integral ideal of norm smaller than 1, which is impossible. Hence the rational field is the only algebraic number field with fundamental discriminant ± 1.

This is the point at which the geometry of numbers enters the ideal theory. The earlier finiteness theorem for classes is sharpened: not only are there finitely many ideal classes, but each class contains an ideal of bounded norm, with a bound expressed solely in terms of the discriminant of the field.

§157. Prime Factors of the Natural Prime Numbers

We now determine how an ordinary rational prime number p decomposes into prime factors in the field Ω. The following treatment follows Hensel's method.

Let $\mathfrak{p}$ be a prime functional, or prime ideal, of degree f. Thus the number of residue classes modulo $\mathfrak{p}$ is p^f, where p is the rational prime lying below $\mathfrak{p}$. The smallest positive rational integer divisible by $\mathfrak{p}$ is then p. Congruences modulo $\mathfrak{p}$ may be applied not only to numbers of the field, but also to forms with rational integral coefficients in auxiliary variables.

Let

$$\tau = \omega_1 t_1 + \omega_2 t_2 + \cdots + \omega_n t_n$$

be the basis form of the field, and let

$$F(z) = N(z - \tau)$$

be the equation satisfied by τ; its coefficients are rational integral forms in $t_1, \ldots, t_n$. Since $F(\tau) = 0$, the congruence $F(\tau) \equiv 0 \pmod{\mathfrak{p}}$ is immediate.

If a rational integral form $\Phi(z)$ satisfies

$$\Phi(\tau) \equiv 0 \pmod{\mathfrak{p}}, \tag{1}$$

then τ may be regarded as a root of the congruence $\Phi(z) \equiv 0 \pmod{\mathfrak{p}}$. In a residue field with p^f elements the Frobenius operation gives new roots. Thus, setting

$$\tau_r = \omega_1^{p^r} t_1 + \omega_2^{p^r} t_2 + \cdots + \omega_n^{p^r} t_n,$$

we have, together with τ,

$$\tau, \tau_1, \ldots, \tau_{f-1}$$

as roots. Moreover $\tau_r \equiv \tau_s \pmod{\mathfrak{p}}$ precisely when $r \equiv s \pmod{f}$, so that these f roots are distinct modulo $\mathfrak{p}$.

It follows that the product

$$P(z) = (z - \tau)(z - \tau_1) \cdots (z - \tau_{f-1}) \tag{2}$$

is congruent modulo $\mathfrak{p}$ to a rational integral form of degree f in $z, t_1, \ldots, t_n$; its coefficients are determined modulo p. We shall denote this form again by $P(z)$.

The form $P(z)$ is irreducible in the following sense. If $\Phi(z)$ is a rational integral form for which $\Phi(\tau)$ is divisible by $\mathfrak{p}$, then

$$\Phi(z) \equiv P(z)\Phi_0(z) \pmod{p} \tag{3}$$

with a rational integral form $\Phi_0(z)$. Thus P has the smallest possible degree in z among all congruences over the rational residue field that are satisfied by τ modulo $\mathfrak{p}$.

We next recover the prime ideal $\mathfrak{p}$ itself from P. The greatest common divisor of the rational prime p and the number $P(\tau)$ is precisely $\mathfrak{p}$. Indeed, $P(\tau)$ is divisible by $\mathfrak{p}$, by construction. If another prime divisor $\mathfrak{q}$ of p also divided $P(\tau)$, then P would vanish at the corresponding residue value of τ, contrary to the irreducibility just established for the factor attached to $\mathfrak{p}$, unless $\mathfrak{q} = \mathfrak{p}$.

Now let

$$p = \mathfrak{p}_1^{e_1}\mathfrak{p}_2^{e_2}\cdots\mathfrak{p}_r^{e_r} \tag{4}$$

be the decomposition of p in Ω, and let f_i be the degree of $\mathfrak{p}_i$. To each $\mathfrak{p}_i$ corresponds a form $P_i(z)$ of degree f_i. Since $F(z) = N(z-\tau)$ is divisible modulo $\mathfrak{p}_i$ by $P_i(z)$, and since every factor over p is obtained in this way, we have the congruence

$$F(z) \equiv P_1(z)^{e_1}P_2(z)^{e_2}\cdots P_r(z)^{e_r} \pmod{p}. \tag{5}$$

The degrees give the familiar relation

$$n = e_1f_1 + e_2f_2 + \cdots + e_rf_r. \tag{6}$$

This is Dedekind's decomposition law in the form required here: the decomposition of the minimal congruence $F(z)$ modulo p determines the prime ideal factors over p, their residue degrees f_i, and their exponents e_i.

§158. Dedekind's Theorem on the Field Discriminant

Let again

$$\tau = \omega_1 t_1 + \cdots + \omega_n t_n$$

be the basis form of Ω, and put

$$F(z) = N(z - \tau).$$

The powers

$$1, \tau, \tau^2, \ldots, \tau^{n-1}$$

may be expressed linearly in the basis $\omega_1, \ldots, \omega_n$. Let U be the determinant of these coefficients. Then the discriminant of the system $1, \tau, \ldots, \tau^{n-1}$ is

$$\Delta(1, \tau, \ldots, \tau^{n-1}) = (-1)^{n(n-1)/2} N(F'(\tau)). \tag{1}$$

On the other hand it is also equal to

$$U^2 \Delta, \tag{2}$$

where Δ is the field discriminant.

It remains to see that U is a unit. If a rational prime p divided U, then the powers $1, \tau, \ldots, \tau^{n-1}$ would be linearly dependent modulo p. There would therefore be a congruence of degree $< n$, with rational integral coefficients, satisfied by τ modulo a prime ideal above p. This contradicts the minimality of the factors P_i established in the preceding section. Hence $U = \pm 1$.

Thus

$$|N(F'(\tau))| = |\Delta|. \tag{3}$$

The number $F'(\tau)$, or more precisely the functional which it determines, is therefore the fundamental functional of the field. Changing the basis form τ only replaces $F'(\tau)$ by an associated functional, so that the ideal so defined is independent of the auxiliary choice.

We now compare this with the decomposition of a rational prime. Suppose that p has the factorization

$$p = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_r^{e_r}$$

and that the factor $P_i(z)$ of $F(z)$ belonging to $\mathfrak{p}_i$ occurs with exponent e_i modulo p. From

$$F(z) \equiv P_i(z)^{e_i}\Phi_i(z) \pmod{p}$$

we get, by differentiation,

$$F'(z) \equiv e_i P_i(z)^{e_i-1}P_i'(z)\Phi_i(z) + P_i(z)^{e_i}\Phi_i'(z) \pmod{p}.$$

Putting $z = \tau$, the factor $P_i(\tau)$ is divisible by $\mathfrak{p}_i$ but not by a higher power than is forced by the exponent already visible in the congruence. Hence the fundamental functional $F'(\tau)$ is divisible by $\mathfrak{p}_i^{e_i-1}$. If e_i is not divisible by p, it is not divisible by $\mathfrak{p}_i^{e_i}$. If e_i is divisible by p, a higher divisibility may occur.

The principal conclusion is the theorem of Dedekind:

A rational prime p divides the field discriminant Δ if and only if, in the field Ω, it is divisible by the square of at least one prime ideal factor. Equivalently, p is ramified in Ω if and only if $p \mid \Delta$.

The formula also gives a useful basis for the inverse different. If

$$F(z) = z^n + a_1 z^{n-1} + \cdots + a_n,$$

and if

$$F_\nu(z) = z^\nu + a_1 z^{\nu-1} + \cdots + a_\nu \qquad (\nu = 0, \ldots, n-1),$$

with $F_0 = 1$, then

$$\frac{F_\nu(\tau)}{F'(\tau)}$$

has trace 0 or 1 according to the usual duality rule. More explicitly, the expansion

$$\frac{F(z)}{(z-\tau)F'(\tau)}$$

has trace 1. Hence the fractions

$$\frac{F_0(\tau)}{F'(\tau)}, \ldots, \frac{F_{n-1}(\tau)}{F'(\tau)}$$

form a system dual to the powers $1, \tau, \ldots, \tau^{n-1}$ with respect to the trace. Therefore, if ω is any whole number of Ω, then

$$\operatorname{Tr}\left(\frac{\omega}{F'(\tau)}\right)$$

is a rational integer. This is the trace-theoretic form of the fundamental ideal.

Nineteenth Section. The Relation of a Field to Its Divisors

§159. Relative Norms

The preceding theory was developed over the rational field. We now let the ground field itself be an algebraic field.

Let R be a field of degree m over the rational field, and let

$$\Omega = R(\Theta)$$

be an algebraic extension of relative degree n. The absolute degree of Ω is then mn. If ρ is a primitive element of R, then the products

$$\rho^i \Theta^j$$

may be used, in the usual range of exponents, to represent the elements of Ω. Every number of Ω is expressible as a polynomial in Θ of degree $< n$, with coefficients in R.

Let

$$\Omega_1, \ldots, \Omega_n$$

be the conjugate fields of Ω over the fixed field R. If $\omega \in \Omega$, its *relative norm* is

$$N_{\Omega/R}(\omega) = \omega_1 \omega_2 \cdots \omega_n, \tag{1}$$

where ω_i denotes the image of ω in Ω_i. This is a number of R. The absolute norm is then obtained by taking the norm in R:

$$N_{\Omega/\mathbb{Q}}(\omega) = N_{R/\mathbb{Q}}\Big(N_{\Omega/R}(\omega)\Big). \tag{2}$$

In exactly the same way one defines the relative norm of a functional or an ideal in Ω; it is then a functional or ideal in R.

Similarly, the relative trace is

$$S_{\Omega/R}(\omega) = \omega_1 + \cdots + \omega_n, \tag{3}$$

and the absolute trace is

$$S_{\Omega/\mathbb{Q}}(\omega) = S_{R/\mathbb{Q}}\Big(S_{\Omega/R}(\omega)\Big). \tag{4}$$

The fundamental ideal is compatible with passage through intermediate fields. Let τ be a basis form of Ω over the rational field, and let

$$F(z) = N_{\Omega/\mathbb{Q}}(z - \tau).$$

Then $F'(\tau)$ represents the absolute fundamental ideal of Ω. Let θ be a basis form of R, with

$$\psi(z) = N_{R/\mathbb{Q}}(z - \theta),$$

and let

$$f(z) = N_{\Omega/R}(z - \tau)$$

be the relative equation of τ over R. The derivative $f'(\tau)$ defines the relative fundamental ideal of Ω over R.

From the chain rule for norms and from the trace-duality interpretation of the derivative one obtains

$$F'(\tau) \sim f'(\tau)\, \psi'(\theta), \tag{5}$$

where $\sim$ denotes association by a unit. Thus

$$\mathfrak{D}_{\Omega/\mathbb{Q}} \sim \mathfrak{D}_{\Omega/R}\, \mathfrak{D}_{R/\mathbb{Q}}. \tag{6}$$

More generally, for a tower of fields the corresponding fundamental ideals multiply transitively.

It is important that the relative fundamental ideal may be a unit even when the absolute fundamental ideal is not. The ramification which is visible over the rational field can disappear after the ground field has been enlarged; the relative ideal records only the ramification still present over R.

We shall call the extension Ω/R normal if every conjugate of Ω over R is again equal to Ω. In that case the substitutions carrying Ω into itself over R form a group, the Galois group of Ω over R. If this group is commutative, the relative field is Abelian.

§160. Prime Ideals in the Relatively Normal Field

Assume now that Ω is normal over R, and denote its group over R by $\mathcal{O}$. If $\varphi \in \mathcal{O}$, and if $\omega \in \Omega$, write

$$\omega^{\varphi}$$

for the conjugate of ω obtained by the substitution φ. The same notation will be used for ideals:

$$\mathfrak{a}^{\varphi} = \{\alpha^{\varphi} : \alpha \in \mathfrak{a}\}.$$

Prime ideals go into prime ideals under the substitutions of the group.

Let $\mathfrak{p}$ be a prime ideal of Ω. There is a unique prime ideal $\mathfrak{P}$ of R below it. Every number of R that is divisible by $\mathfrak{p}$ is divisible by $\mathfrak{P}$, and conversely the ideal $\mathfrak{P}$ in Ω is divisible by all conjugates of $\mathfrak{p}$. If

$$N_{\Omega/R}(\mathfrak{p}) = \mathfrak{P}^{f}, \tag{1}$$

the exponent f is called the degree of $\mathfrak{p}$ over R.

Let the distinct conjugates of $\mathfrak{p}$ under the group be

$$\mathfrak{p}_1, \mathfrak{p}_2, \dots, \mathfrak{p}_g.$$

Suppose that the highest power of each of them dividing $\mathfrak{P}$ is the e-th power. Then

$$\mathfrak{P} = (\mathfrak{p}_1\mathfrak{p}_2\cdots\mathfrak{p}_g)^{e}. \tag{2}$$

Taking relative norms gives the fundamental relation

$$n = efg, \tag{3}$$

where $n = [\Omega : R]$.

The substitutions that carry $\mathfrak{p}$ into itself form a subgroup Ψ of $\mathcal{O}$. This is the decomposition group of $\mathfrak{p}$. Its cosets describe the conjugate prime ideals. The substitutions in Ψ act on the residue field $o_{\Omega}/\mathfrak{p}$, and their action fixes the residue field of R modulo $\mathfrak{P}$. The quotient by the subgroup acting trivially will therefore measure the residue degree.

The preceding factorization is the normal-field form of the decomposition law. In a non-normal field one must pass to a normal overfield, or else use the double-coset formula described later. In the normal case the symmetry supplied by $\mathcal{O}$ ensures that all prime ideals over a fixed $\mathfrak{P}$ have the same ramification exponent e and the same residue degree f.

§161. Primitive Roots of Prime Ideals

Let $\mathfrak{P}$ be a prime ideal of R, and let $\mathfrak{p}$ be a prime ideal of Ω above it, of degree f. Denote by $P = N_{R/\mathbb{Q}}(\mathfrak{P})$ the number of residue classes modulo $\mathfrak{P}$ in R. Then the number of residue classes modulo $\mathfrak{p}$ in Ω is

$$P^f.$$

Consequently every number α of Ω satisfies

$$\alpha^{P^f} \equiv \alpha \pmod{\mathfrak{p}}. \tag{1}$$

If α is not divisible by $\mathfrak{p}$, then

$$\alpha^{P^f-1} \equiv 1 \pmod{\mathfrak{p}}.$$

As in the rational case, the non-zero residue classes modulo $\mathfrak{p}$ form a cyclic group. There is therefore a residue class γ, represented by a whole number of Ω, such that

$$\gamma^{P^f-1} \equiv 1 \pmod{\mathfrak{p}}$$

and no smaller positive exponent gives 1. The powers

$$1, \gamma, \gamma^2, \ldots, \gamma^{P^f-2}$$

then represent all non-zero residue classes modulo $\mathfrak{p}$. Such a γ will be called a primitive root of the prime ideal $\mathfrak{p}$.

Let θ be any whole number of Ω. Among all congruences over R satisfied by θ modulo $\mathfrak{p}$, choose one of least degree:

$$G(\theta) \equiv 0 \pmod{\mathfrak{p}}. \tag{2}$$

Division by powers of γ shows that every residue class modulo $\mathfrak{p}$ is representable by a polynomial in γ with coefficients taken modulo $\mathfrak{P}$. Hence the degree of G cannot be less than f.

On the other hand, the f elements

$$\gamma, \gamma^P, \gamma^{P^2}, \ldots, \gamma^{P^{f-1}}$$

are distinct modulo $\mathfrak{p}$, and all are roots of the same congruence over R. Therefore

$$(z-\gamma)(z-\gamma^P)\cdots(z-\gamma^{P^{f-1}}) \tag{3}$$

is congruent modulo $\mathfrak{p}$ to a whole polynomial $G(z)$ of degree f with coefficients in R. This polynomial is the relative irreducible congruence belonging to the prime ideal $\mathfrak{p}$. It plays, over R, the same role as the polynomial $P(z)$ of §157 played over the rational field.

The construction also shows that the residue field modulo $\mathfrak{p}$ is obtained from the residue field modulo $\mathfrak{P}$ by adjoining the residue class of γ. Its degree is exactly f.

§162. The Partial Fundamental Ideal

Let τ be a basis form of Ω over R, and let $\mathfrak{p}$ be a prime ideal of Ω above $\mathfrak{P}$. Among the substitutions in the decomposition group Ψ of $\mathfrak{p}$, consider first those for which

$$\tau^{\chi} \equiv \tau \pmod{\mathfrak{p}}. \tag{1}$$

They form a subgroup X of Ψ. This subgroup is the inertia group. Its elements act trivially on the residue field.

There is also a distinguished coset in Ψ/X, represented by a substitution ψ_0, characterized by

$$\omega^{\psi_0} \equiv \omega^{P} \pmod{\mathfrak{p}} \tag{2}$$

for every whole number ω of Ω. In other words, ψ_0 induces the Frobenius substitution on the residue field. The decomposition group splits into cosets

$$\Psi = X + X\psi_0 + \cdots + X\psi_0^{f-1}. \tag{3}$$

The quotient Ψ/X is cyclic of order f.

The subgroup X may itself have a filtration by higher congruence conditions. Let $X^{(1)}$ consist of those substitutions $\chi \in X$ for which

$$\tau^{\chi} \equiv \tau \pmod{\mathfrak{p}^2};$$

let $X^{(2)}$ consist of those for which the congruence holds modulo $\mathfrak{p}^3$, and so on. We obtain a chain

$$X \supset X^{(1)} \supset X^{(2)} \supset \cdots, \tag{4}$$

which eventually reaches the identity. Hilbert's terminology is: Ψ is the decomposition group, X the inertia group, and $X^{(1)}$ the ramification group.

We now compute the contribution of $\mathfrak{p}$ to the relative fundamental ideal. If

$$f(z) = N_{\Omega/R}(z - \tau),$$

then the relative fundamental ideal is represented by

$$f'(\tau) = \prod_{\varphi \neq 1} (\tau - \tau^{\varphi}). \tag{5}$$

The factor $\tau - \tau^{\varphi}$ is divisible by $\mathfrak{p}$ precisely when $\varphi \in X$, except for the identity factor which is omitted. It is divisible by $\mathfrak{p}^2$ precisely when $\varphi \in X^{(1)}$, by $\mathfrak{p}^3$ precisely when $\varphi \in X^{(2)}$, and so forth. If

$$|X| = e, \quad |X^{(1)}| = e_1, \quad |X^{(2)}| = e_2, \dots,$$

then the exponent with which $\mathfrak{p}$ enters the relative fundamental ideal is

$$(e-1) + (e_1 - 1) + (e_2 - 1) + \cdots . \tag{6}$$

The same exponent occurs for every conjugate of $\mathfrak{p}$.

Consequently, if $\mathfrak{p}_1, \dots, \mathfrak{p}_g$ are the prime ideals over $\mathfrak{P}$, the relative fundamental ideal contains the factor

$$(\mathfrak{p}_1 \mathfrak{p}_2 \cdots \mathfrak{p}_g)^{(e-1)+(e_1-1)+(e_2-1)+\cdots}. \tag{7}$$

Taking the norm down to R gives the relative discriminant exponent. For $R = \mathbb{Q}$ this formula becomes the usual discriminant formula for the field Ω.

§163. The Ideals in the Divisor Fields of Ω

Let Ω/R be normal with group $\mathcal{O}$, and let $\mathcal{O}'$ be a subgroup. The elements fixed by $\mathcal{O}'$ form a divisor field Ω' of Ω. If

$$|\mathcal{O}'| = n',$$

and if $|\mathcal{O}| = n$, then

$$[\Omega : \Omega'] = n', \qquad [\Omega' : R] = \frac{n}{n'}.$$

The extension Ω/Ω' is normal, though Ω'/R need not be normal.

We describe the prime ideals of Ω' lying over a given prime ideal $\mathfrak{P}$ of R. Let $\mathfrak{p}$ be a prime ideal of Ω over $\mathfrak{P}$, with decomposition group Ψ. The prime ideals in Ω' are obtained by grouping the conjugates of $\mathfrak{p}$ under the action of $\mathcal{O}'$. This grouping is governed by double cosets:

$$\mathcal{O} = \Psi\varphi_1\mathcal{O}' + \Psi\varphi_2\mathcal{O}' + \cdots + \Psi\varphi_s\mathcal{O}'. \tag{1}$$

Each double coset gives one prime ideal $\mathfrak{p}'_r$ of Ω' over $\mathfrak{P}$.

Let $\Psi_r = \varphi_r^{-1}\Psi\varphi_r$, and let

$$H_r = \Psi_r \cap \mathcal{O}'.$$

This group is the decomposition group, inside Ω/Ω', of a prime ideal of Ω lying over $\mathfrak{p}'_r$. Its inertia subgroup and residue degree determine how $\mathfrak{p}'_r$ decomposes further in Ω.

Thus the factorization of $\mathfrak{P}$ in Ω' has the form

$$\mathfrak{P} = (\mathfrak{p}'_1)^{a_1}(\mathfrak{p}'_2)^{a_2}\cdots(\mathfrak{p}'_s)^{a_s}. \tag{2}$$

The exponents a_r are obtained from the inertia indices supplied by the intersections H_r. Their sum, with residue degrees counted, gives the degree

$$[\Omega' : R].$$

When $\mathfrak{p}'_r$ is further decomposed in Ω, say

$$\mathfrak{p}'_r = (\mathfrak{p}_{r1}\mathfrak{p}_{r2}\cdots\mathfrak{p}_{r,e_r})^{g_r}, \tag{3}$$

the relative norm has the form

$$N_{\Omega/\Omega'}(\mathfrak{p}_{rs}) = (\mathfrak{p}'_r)^{f_r}, \tag{4}$$

and

$$n' = e_r f_r g_r. \tag{5}$$

If f is the degree of $\mathfrak{p}$ over R, and f'_r is the degree of $\mathfrak{p}'_r$ over R, then

$$f = f_r f'_r. \tag{6}$$

This identity expresses the transitivity of residue degrees in the tower

$$\Omega \supset \Omega' \supset R.$$

The complete law of decomposition in an intermediate field is therefore the double-coset law: double cosets of the decomposition group of $\mathfrak{p}$ with the subgroup defining the intermediate field correspond to the prime ideals in that intermediate field, and the intersections determine ramification and residue degree.

Twentieth Section. Quadratic Fields

§164. Basis of a Quadratic Field

Let

$$\Omega = \mathbb{Q}(\sqrt{d})$$

be a quadratic field, where d is a square-free rational integer different from 1. If $d > 0$ the field is real; if $d < 0$ it is imaginary. The field is normal, since the only non-trivial substitution changes $\sqrt{d}$ into $-\sqrt{d}$.

Every number of the field is of the form

$$x + y\sqrt{d}, \qquad x, y \in \mathbb{Q}.$$

If ω is an irrational number of the field, then it satisfies an equation

$$c\omega^2 = a + b\omega \tag{1}$$

with rational integers a, b, c. The discriminant of this equation is

$$D = b^2 + 4ac,$$

and D/d is a square. Conversely the roots are

$$\omega = \frac{b \pm \sqrt{D}}{2c}. \tag{2}$$

A number is integral if its equation may be taken monic. From (2) one obtains the usual distinction according to the residue of d modulo 4. If

$$d \equiv 2, 3 \pmod 4,$$

then the whole numbers of Ω are precisely

$$x + y\sqrt{d}, \qquad x, y \in \mathbb{Z}. \tag{3}$$

If

$$d \equiv 1 \pmod 4,$$

then the whole numbers are

$$\frac{x + y\sqrt{d}}{2},$$

where x and y are both even or both odd. Equivalently they are

$$x + y\theta, \qquad \theta = \frac{1 + \sqrt{d}}{2}, \quad x, y \in \mathbb{Z}. \tag{4}$$

Thus a basis of the ring of integers is

$$1, \sqrt{d} \qquad (d \equiv 2, 3 \pmod 4), \tag{5a}$$

and

$$1, \frac{1+\sqrt{d}}{2} \qquad (d \equiv 1 \pmod 4). \tag{5b}$$

It is convenient to write in both cases

$$1, \theta$$

for the basis, with

$$\theta = \sqrt{d}$$

in the first case and

$$\theta = \frac{1+\sqrt{d}}{2}$$

in the second.

The discriminant, which Weber calls the ground number, is therefore

$$\Delta = 4d \qquad (d \equiv 2, 3 \pmod 4), \tag{6a}$$

and

$$\Delta = d \qquad (d \equiv 1 \pmod 4). \tag{6b}$$

If θ' is the conjugate of θ, then

$$\theta - \theta' = \sqrt{\Delta} \tag{7}$$

with the evident choice of the square-root. This compact formula is useful in the following sections.

The values $\Delta = \pm 1, \pm 2$ cannot occur as ground numbers of quadratic fields. The smallest absolute value of a negative ground number is therefore 3, attained by the field $\mathbb{Q}(\sqrt{-3})$.

§165. Prime Ideals of the First Degree

Let φ be a whole functional in the quadratic field. Its basis may be written in the form

$$\alpha_1 = a_{11}, \qquad \alpha_2 = a_{12} + a_{22}\theta, \tag{1}$$

where a_{11} is the smallest positive rational integer divisible by φ, and a_{22} is the least positive rational integer for which $a_{22}\theta$ is congruent modulo φ to a rational integer. The absolute norm of φ is

$$N_a(\varphi) = a_{11}a_{22}. \tag{2}$$

Apply this to a prime functional π. Since the field has degree 2, the prime ideal π can have degree 1 or 2. If its absolute norm is p, then the rational prime p decomposes into two prime factors. If its absolute norm is p^2, then p remains prime in the quadratic field.

For $\varphi = \pi$ one has at once

$$a_{11} = p.$$

The number a_{22} is determined by the condition that $a_{22}\theta$ be congruent to a rational number modulo π. The necessary and sufficient condition is

$$a_{22}(\theta^p - \theta) \equiv 0 \pmod{\pi}. \tag{3}$$

If

$$\theta^p - \theta \equiv 0 \pmod{\pi},$$

then $a_{22} = 1$, and the prime ideal has degree 1. Otherwise $a_{22} = p$, and the prime ideal has degree 2.

In the first case θ, and therefore every whole number of Ω, is congruent modulo π to a rational number. The p rational residues

$$0, 1, \ldots, p-1$$

then form a complete residue system modulo π. Thus

$$N_a(\pi) = p, \qquad a_{22} = 1. \tag{4a}$$

In the second case

$$N_a(\pi) = p^2, \qquad a_{22} = p, \tag{4b}$$

and the prime functional is principal:

$$\pi = p(u + \theta v). \tag{5}$$

Suppose now that p splits. Then a_{12} is determined by

$$a_{12} + \theta \equiv 0 \pmod{\pi}. \tag{6}$$

Taking the norm, the product

$$(a_{12} + \theta)(a_{12} + \theta')$$

must be divisible by p. Using

$$\theta - \theta' = \sqrt{\Delta},$$

one obtains the following criterion. The rational prime p splits into two prime factors in the quadratic field if and only if the congruence

$$x^2 \equiv \Delta \pmod{4p} \tag{7}$$

is soluble in rational integers x. In Weber's terminology, the ground number Δ must be a quadratic residue modulo $4p$.

If x is chosen so that (7) holds, the two conjugate prime functionals may be written

$$\pi = pu + (a_{12} + \theta)v, \qquad \pi' = pu + (a_{12} + \theta')v. \tag{8}$$

Their product is associated with the rational prime p.

Finally, the two conjugate factors π, π' may themselves be associated. This happens exactly when π', hence also $\pi - \pi'$, is divisible by π. Since

$$\pi - \pi' = (\theta - \theta')v,$$

the condition is that

$$(\theta - \theta')^2 = \Delta$$

be divisible by p. Thus p is associated with the square of a prime functional only when $p \mid \Delta$, in agreement with the general ramification theorem of §158.

§166. Functionals in the Quadratic Field and Quadratic Irrationalities

We now take an arbitrary whole functional φ of the quadratic field and write its basis in the form

$$\alpha_1 = a_{11}, \qquad \alpha_2 = a_{12} + a_{22}\theta. \tag{1}$$

Then

$$N_a(\varphi) = a_{11}a_{22}. \tag{2}$$

Put

$$\varphi = a_{11}t_1 + (a_{12} + a_{22}\theta)t_2. \tag{3}$$

Taking the norm of this linear form gives a binary quadratic form. Its divisor is $a_{11}a_{22}$. We write

$$a_{11} = c\, a_{11}a_{22},$$

and, after dividing the coefficients of the norm form by the common divisor, obtain relatively prime rational integers a, b, c for which the quotient

$$\omega = \frac{-b + \sqrt{\Delta}}{2c} \tag{4}$$

satisfies

$$c\omega^2 = a + b\omega, \tag{5}$$

or equivalently

$$b^2 + 4ac = \Delta. \tag{6}$$

Thus ω is a quadratic irrationality belonging to the discriminant Δ.

The functional φ itself may then be written

$$\varphi = ec(t_1 + \omega t_2), \tag{7}$$

where e is a positive rational integer. Here c is the least positive integer for which $c\omega$ is a whole number of the quadratic field; every integer h with $h\omega$ integral is divisible by c.

Conversely, let

$$\omega = \frac{-b + \sqrt{\Delta}}{2c}$$

be any quadratic irrationality of discriminant Δ, with a, b, c relatively prime and satisfying $b^2 + 4ac = \Delta$. Then for every positive integer e the form

$$ec(t_1 + \omega t_2) \tag{8}$$

is the basis form of a whole functional. Indeed its coefficients ec and $ec\omega$ are whole numbers, and the norm calculation gives

$$N_a(\varphi) = e^2 c. \tag{9}$$

The least rational integer divisible by the functional is ec, and the second basis element is $ec\omega$.

The result may be stated succinctly.

Every whole functional φ in a quadratic field with ground number Δ determines a quadratic irrationality ω of discriminant Δ and a positive integer e such that

$$ec(t_1 + \omega t_2) \tag{10}$$

is a basis form of φ. Conversely, every quadratic irrationality of discriminant Δ, together with a positive integer e, gives in this way a whole functional of the quadratic field.

This is the bridge between the ideal theory of the quadratic field and the classical theory of binary quadratic forms.

§167. Equivalent Functionals and Equivalent Numbers

Let

$$\omega, \omega_1$$

be two quadratic irrationalities belonging to the same discriminant Δ. Their associated basis forms are

$$\varphi = ec(t_1 + \omega t_2), \qquad \varphi_1 = e_1 c_1(t_1 + \omega_1 t_2). \tag{1}$$

Suppose first that the two irrationalities are equivalent in the sense of the theory of binary quadratic forms; that is,

$$\omega_1 = \frac{p\omega + q}{r\omega + s}, \tag{2}$$

where p, q, r, s are rational integers and

$$ps - qr = \pm 1. \tag{3}$$

Substituting (2) into φ_1, and putting

$$u_1 = st_1 + qt_2, \qquad u_2 = rt_1 + pt_2,$$

we obtain

$$(r\omega + s)\varphi_1 = e_1 c_1(u_1 + \omega u_2). \tag{4}$$

The change of variables has determinant ± 1. Hence the form on the right is associated with φ, and therefore the two functionals φ, φ_1 are equivalent.

Conversely, assume that the functionals φ, φ_1 are equivalent. Let ψ be a functional in the reciprocal class to that of φ, so that $\varphi\psi$ is associated with a principal functional. Write

$$\varphi\psi = \varepsilon\xi, \tag{5}$$

where ε is a unit and ξ a number of the ring. Take basis forms

$$\varphi = ec(t_1 + \omega t_2), \qquad \psi = e'c'(t_1 + \eta t_2). \tag{6}$$

Multiplying by the conjugate form of ψ, and comparing discriminants, gives a new basis of ψ'. This basis differs from

$$e'c', \quad e'c'\eta'$$

by an integral unimodular substitution. Therefore there are rational integers p, q, r, s with

$$ps - qr = \pm 1 \tag{7}$$

such that

$$\omega = \frac{p\eta' + q}{r\eta' + s}. \tag{8}$$

Thus ω is equivalent to η'. Replacing φ by an equivalent form φ_1 gives the same ψ, and hence ω_1 is also equivalent to η'. Therefore ω and ω_1 are equivalent.

We have therefore proved the two directions:

If two quadratic irrationalities of discriminant Δ are equivalent, then the corresponding functionals are equivalent; and if two whole functionals are equivalent, then the corresponding quadratic irrationalities are equivalent.

Consequently each ideal class of the quadratic field corresponds to exactly one class of equivalent quadratic irrationalities of discriminant Δ, and conversely. The number of inequivalent quadratic irrationalities of the discriminant Δ is equal to the number of ideal classes of the field.

In the terminology of Gauss one sometimes obtains a finer division into proper and improper equivalence. With the present notion of equivalence of irrationalities, in which this distinction is not made, no such further division is necessary.

§168. Composition of Quadratic Irrationalities

We have seen that to every whole functional φ in the quadratic field there belongs a quadratic irrationality ω of discriminant Δ. We now determine how far ω is fixed by φ, and then define composition.

Let the basis of φ be

$$\alpha_1 = a_{11}, \qquad \alpha_2 = a_{12} + a_{22}\theta. \tag{1}$$

The numbers a_{11} and a_{22} are uniquely determined by φ, since

$$N_a(\varphi) = a_{11}a_{22}. \tag{2}$$

The remaining coefficient a_{12} is determined modulo φ by

$$a_{12} + a_{22}\theta \equiv 0 \pmod{\varphi}. \tag{3}$$

Since a_{12} is rational integral, it is determined only up to a multiple of a_{11}. Replacing

$$\alpha_2$$

by

$$\alpha_2 + h\alpha_1$$

therefore replaces the corresponding irrationality $\omega = \alpha_2/\alpha_1$ by

$$\omega + h.$$

We have the following statement.

To each whole functional of the quadratic field belongs one and only one series of quadratic irrationalities of discriminant Δ. The numbers of this series differ from one another by arbitrary rational integers and are consequently equivalent. Associated functionals lead to the same series.

Now let φ_1, φ_2 be two whole functionals, and put

$$\varphi_3 = \varphi_1\varphi_2. \tag{4}$$

Choose irrationalities $\omega_1, \omega_2, \omega_3$ belonging to $\varphi_1, \varphi_2, \varphi_3$. The class of ω_3 is determined by the classes of ω_1 and ω_2, up to the addition of a rational integer. We call ω_3 the composite of ω_1 and ω_2, and write symbolically

$$\omega_1\omega_2 = \omega_3. \tag{5}$$

If either factor is replaced by an equivalent irrationality, the composite is replaced by an equivalent irrationality. Thus the classes of quadratic irrationalities of the discriminant Δ form an Abelian group, isomorphic to the group of ideal classes of the quadratic field.

It remains to describe the actual computation. Let the bases of the three functionals be

$$\begin{aligned} \varphi_1: &\quad \alpha_1 = a_{11}, \qquad \alpha_2 = a_{12} + a_{22}\theta, \\ \varphi_2: &\quad \beta_1 = b_{11}, \qquad \beta_2 = b_{12} + b_{22}\theta, \\ \varphi_3: &\quad \gamma_1 = c_{11}, \qquad \gamma_2 = c_{12} + c_{22}\theta. \end{aligned} \tag{6}$$

Then

$$\omega_1 = \frac{\alpha_2}{\alpha_1}, \qquad \omega_2 = \frac{\beta_2}{\beta_1}, \qquad \omega_3 = \frac{\gamma_2}{\gamma_1}.$$

The number c_{11} is the least positive rational integer divisible by the product $\varphi_1\varphi_2$, and therefore divides $a_{11}b_{11}$. Write

$$a_{11}b_{11} = kc_{11}. \tag{7}$$

Since absolute norms multiply, one has

$$c_{22} = k\,a_{22}b_{22}. \tag{8}$$

The last coefficient c_{12} is determined by

$$c_{12} + c_{22}\theta \equiv 0 \pmod{\varphi_1\varphi_2}. \tag{9}$$

For the purpose of finding a representative of the composed class one may proceed more simply. Keep φ_1 fixed, and replace φ_2 by an equivalent functional chosen so that it is relatively prime to a_{11}. This can always be done. In that situation b_{11} is relatively prime to a_{11}, and therefore

$$c_{11} = a_{11}b_{11}, \qquad c_{22} = a_{22}b_{22}. \tag{10}$$

The congruence (9) splits into the two congruences

$$\begin{aligned} c_{12} &\equiv b_{22}a_{12} \pmod{a_{11}}, \\ c_{12} &\equiv a_{22}b_{12} \pmod{b_{11}}. \end{aligned} \tag{11}$$

These determine c_{12} modulo $c_{11} = a_{11}b_{11}$. The irrationality

$$\omega_3 = \frac{c_{12} + c_{22}\theta}{c_{11}} \tag{12}$$

is then a representative of the composed class.

It remains only to justify the preliminary replacement of φ_2. Since associated forms are equivalent, φ_2 may be replaced by a form

$$c_2(t_1 + \omega_2 t_2).$$

By making a unimodular substitution

$$\omega_2 \mapsto \frac{p\omega_2 + q}{r\omega_2 + s}, \qquad ps - qr = \pm 1,$$

the leading coefficient c_2 is transformed into

$$c_2' = -a_2 r^2 + b_2 rs + c_2 s^2.$$

The integers r, s may be chosen relatively prime and so that c_2' is not divisible by any prescribed finite set of rational primes. In particular it may be chosen prime to a_{11}. This proves the asserted reduction.

The laws of composition have here been derived under the assumption that the discriminant Δ is the ground number of a quadratic field, that is, a fundamental discriminant. With slight modifications the same laws remain valid for arbitrary discriminants, as in Gauss's theory of binary quadratic forms.

Twenty-First Section. Cyclotomic Fields

§169. Decomposition of the Prime q in the Cyclotomic Field Ω_m

We make a second application of the general theory of algebraic numbers to cyclotomic fields. This will supply the means for bringing the theory of Abelian number fields to a particularly transparent conclusion.

Let q be a rational prime, including the case $q = 2$, and let

$$m = q^\lambda \tag{1}$$

be a positive power of q. In the case $q = 2$ we assume $\lambda > 1$. Let ζ be a primitive m-th root of unity, and denote by Ω_m the full cyclotomic field generated by ζ. Its degree is

$$\mu = \varphi(m) = q^{\lambda-1}(q-1). \tag{2}$$

We let n run through a complete system of residues modulo m, with the residues divisible by q omitted. Then the μ numbers

$$\zeta^n$$

are the roots of the irreducible cyclotomic equation $f(x) = 0$, where

$$f(x) = \frac{x^m - 1}{x^{m/q} - 1}. \tag{3}$$

The root ζ is a whole number, and its norm is $+1$; hence it is a unit.

The field Ω_m is normal. The substitutions

$$s_n = (\zeta, \zeta^n) \tag{4}$$

where n is prime to m, form an Abelian group isomorphic to the group of reduced residue classes modulo m.

Putting $x = 1$ in the factorization

$$f(x) = \prod_n (x - \zeta^n),$$

we get

$$q = \prod_n (1 - \zeta^n). \tag{5}$$

Let

$$\delta_n = 1 - \zeta^n, \qquad \delta = 1 - \zeta. \tag{6}$$

If n, n' are prime to m, then $n' \equiv an \pmod{m}$ for some a, and

$$\frac{1-\zeta^{n'}}{1-\zeta^n} = 1 + \zeta^n + \zeta^{2n} + \cdots + \zeta^{(a-1)n}$$

is a whole number. The reverse divisibility is obtained by interchanging n and n'. Thus all δ_n are associated. Formula (5) therefore gives

$$q \sim \delta^\mu, \qquad N(\delta) = q. \tag{7}$$

Since any proper divisor of δ would have norm dividing q, it follows that δ itself is a prime factor of first degree.

Thus the rational prime q is associated in Ω_m with the μ-th power of a prime factor of the first degree.

If $m = m_1 m_2$, with m_1, m_2 again powers of q, then applying the factorization of $x^{m_1} - 1$ and putting $x = \zeta^{m_2}$ gives

$$N(\zeta^{m_2} - 1) = q^{\varphi(m_1)}. \tag{8}$$

From this one computes the discriminant of the cyclotomic equation. Since

$$\operatorname{disc} f = (-1)^{\mu(\mu-1)/2} N(f'(\zeta)),$$

the discriminant is, apart from sign, a power of q. More explicitly it is the usual cyclotomic discriminant

$$\Delta_m = \pm\, q^{\,q^{\lambda-1}(\lambda q - \lambda - 1)}. \tag{9}$$

Only the prime q occurs in it, as expected from the complete ramification of q in Ω_m.

§170. Minimal Basis of the Field Ω_m

The preceding results lead to the following theorem.

The numbers

$$1, \zeta, \zeta^2, \dots, \zeta^{\mu-1} \tag{1}$$

form a basis of the system o of whole numbers of the cyclotomic field Ω_m.

It is enough to show that a number

$$\omega = x_0 + x_1\zeta + \cdots + x_{\mu-1}\zeta^{\mu-1} \tag{2}$$

with rational integral coefficients can be divisible by a rational prime p only if all the x_i are divisible by p. If ω is divisible by p, then every conjugate

$$\omega_n = x_0 + x_1\zeta^n + x_2\zeta^{2n} + \cdots + x_{\mu-1}\zeta^{(\mu-1)n} \tag{3}$$

is divisible by p. The determinant of this system in the unknowns x_i is the square-root of the discriminant. If $p \neq q$, the discriminant is not divisible by p, and hence all x_i are divisible by p.

It remains to consider $p = q$. Put

$$F(t) = x_0 + x_1 t + \cdots + x_{\mu-1}t^{\mu-1}.$$

Since $\zeta = 1 - \delta$, Taylor's expansion gives

$$\omega = F(1-\delta) = F(1) - \delta F'(1) + \frac{\delta^2}{2!}F''(1) - \cdots . \tag{4}$$

If ω is divisible by q, then it is divisible by δ^μ. From (4) it follows successively that

$$F(1), F'(1), F''(1), \dots$$

are divisible by q. Written out in terms of the x_i, these congruences imply successively that

$$x_{\mu-1}, x_{\mu-2}, \dots, x_0$$

are all divisible by q. This proves the theorem.

Instead of the sequence (1), one may take any sequence of μ consecutive powers of ζ,

$$\zeta^a, \zeta^{a+1}, \dots, \zeta^{a+\mu-1}, \tag{5}$$

as a basis. This follows by multiplication with the unit ζ^a. In particular, the symmetric sequence of powers around 1 is often useful.

The ground number of the field is therefore the discriminant of the cyclotomic equation itself:

$$\Delta(\Omega_m) = \mathrm{disc}(1, \zeta, \dots, \zeta^{\mu-1}). \tag{6}$$

Thus the discriminant computed in the preceding section is the field discriminant.

§171. The Prime Ideals in the Field Ω_m

We have already seen that the prime q belonging to the exponent $m = q^\lambda$ is totally ramified:

$$q \sim (1-\zeta)^\mu.$$

It remains to decompose the rational primes $p \neq q$.

Let p be such a prime. If

$$\omega = x_0 + x_1\zeta + \cdots + x_{\mu-1}\zeta^{\mu-1} \tag{1}$$

is a whole number, then its conjugates are

$$\omega_n = x_0 + x_1\zeta^n + x_2\zeta^{2n} + \cdots + x_{\mu-1}\zeta^{(\mu-1)n}. \tag{2}$$

Two such conjugates can be congruent modulo a prime ideal $\mathfrak{p}$ over p only when the corresponding exponents are congruent modulo m. Hence the inertia group is trivial: no prime $p \neq q$ is divisible by the square of a prime ideal in Ω_m. This is also clear from the discriminant, which is a power of q.

Raising (2) to the p-th power and using Fermat's theorem for the rational coefficients gives

$$\omega^p \equiv \omega_p \pmod{\mathfrak{p}}. \tag{3}$$

Thus the Frobenius substitution belonging to $\mathfrak{p}$ is

$$\zeta \mapsto \zeta^p.$$

Let f be the least positive exponent for which

$$p^f \equiv 1 \pmod{m}. \tag{4}$$

Then the decomposition group is generated by

$$(\zeta, \zeta^p),$$

and has order f. Hence every prime ideal over p has residue degree f, and

$$N(\mathfrak{p}) = p^f. \tag{5}$$

If

$$\mu = \varphi(m) = ef,$$

then there are e distinct conjugate prime ideals over p. Therefore:

If $p \neq q$ belongs to the exponent f modulo m, and if $\varphi(m) = ef$, then p decomposes in Ω_m into e distinct conjugate prime factors, each of degree f.

§172. The Conjugate Prime Ideals

We now describe the conjugate prime ideals over a prime $p \neq q$ more explicitly. Let again f be the least positive exponent with

$$p^f \equiv 1 \pmod{m}, \qquad \varphi(m) = ef. \tag{1}$$

The subgroup of the residue-class group modulo m generated by p is

$$1, p, p^2, \ldots, p^{f-1}. \tag{2}$$

Choose representatives

$$\xi_1, \xi_2, \ldots, \xi_e$$

for the cosets of this subgroup in the group of reduced residue classes modulo m. Then the prime factors over p may be denoted

$$\mathfrak{p}_{\xi_1}, \mathfrak{p}_{\xi_2}, \ldots, \mathfrak{p}_{\xi_e},$$

and one has, up to association,

$$p = \mathfrak{p}_{\xi_1}\mathfrak{p}_{\xi_2} \cdots \mathfrak{p}_{\xi_e}. \tag{3}$$

For an odd $m = q^\lambda$, let g be a primitive root modulo m, and write

$$p \equiv g^\gamma \pmod{m}.$$

Then f is the least positive integer with

$$\gamma f \equiv 0 \pmod{\varphi(m)},$$

and $e = (\gamma, \varphi(m))$. The coset representatives may be taken as

$$1, g, g^2, \ldots, g^{e-1}. \tag{4}$$

If $p-1$ is divisible by q, a part of the splitting already occurs in the smaller cyclotomic field determined by the greatest common divisor of $p-1$ and m. If $p-1$ is not divisible by q, then the series (4) gives the representatives directly.

For $m = 2^\lambda$ the same discussion is made with the customary generators of the odd residue classes; if $p-1$ is divisible by 4, powers of 5 are used, and if only divisibility by 2 occurs, the representatives are correspondingly adjusted. The exceptional small case $m = 4$ is to be separated.

The substance of the result is that the decomposition of p is completely governed by the order of p in the multiplicative group of residues modulo m. The conjugate prime ideals correspond to the cosets of the subgroup generated by p.

§173. Representation of the Prime Factors of p

The representation of the prime factors can now be made quite explicit. Let

$$f(t) = \prod_n (t - \zeta^n)$$

be the cyclotomic polynomial, the product being over the reduced residues n modulo m. For a fixed prime $p \neq q$, with order f modulo m, put

$$F_n(t) = \prod_{h=0}^{f-1} (t - \zeta^{np^h}). \tag{1}$$

This depends only on the coset of n modulo the subgroup generated by p. If $\xi_1, \ldots, \xi_e$ are coset representatives, then

$$f(t) = F_{\xi_1}(t) F_{\xi_2}(t) \cdots F_{\xi_e}(t). \tag{2}$$

By the binomial theorem,

$$(t - \zeta^{np^h})^p \equiv t^p - \zeta^{np^{h+1}} \pmod{p},$$

and the exponents p^{h+1} run through the same cycle as the exponents p^h. Hence

$$F_n(t)^p \equiv F_n(t^p) \pmod{p}. \tag{3}$$

This is exactly the criterion that $F_n(t)$ be congruent modulo p to a polynomial $P_n(t)$ with rational integral coefficients:

$$F_n(t) \equiv P_n(t) \pmod{p}. \tag{4}$$

Accordingly,

$$f(t) \equiv P_{\xi_1}(t) P_{\xi_2}(t) \cdots P_{\xi_e}(t) \pmod{p}. \tag{5}$$

The prime ideal $\mathfrak{p}_1$ corresponding to the coset of 1 is the greatest common divisor of p and

$$P_1(\zeta). \tag{6}$$

More generally, $\mathfrak{p}_n$ is the greatest common divisor of p and $P_1(\zeta^n)$, or, if $nn' \equiv 1 \pmod{m}$, the greatest common divisor of p and $P_{n'}(\zeta)$. Thus all prime factors of p are obtained by taking the greatest common divisors of p with the functions

$$P_{\xi_1}(\zeta), \ldots, P_{\xi_e}(\zeta). \tag{7}$$

In the special case

$$p \equiv 1 \pmod{m}, \tag{8}$$

we have $f = 1$ and $e = \varphi(m)$. Then every factor is of the first degree. The root ζ is congruent modulo each prime ideal $\mathfrak{p}_n$ to a rational residue. If g is a primitive root modulo p, this may be written in the form

$$\zeta \equiv g^{(p-1)n/m} \pmod{\mathfrak{p}_n}, \tag{9}$$

with n prime to m. Consequently $\mathfrak{p}_n$ is the greatest common divisor of p and

$$\zeta - g^{(p-1)n'/m}, \tag{10}$$

where

$$nn' \equiv 1 \pmod{m}. \tag{11}$$

This gives an exact characterization of each prime ideal in the completely split case.

We shall also use the following terminology. In the factorization of any number or functional into prime factors, a prime ideal $\mathfrak{p}$ occurs with a well-defined exponent, positive, negative, or zero. If the exponent is zero, $\mathfrak{p}$ is not contained in the number. Two numbers or functionals are called related if they contain a common prime functional, and relatively prime if they contain none. A number which has no related rational prime is a unit. In a normal field all conjugates of a number are related to the same rational primes.

§174. Kummer's Theorem

We now return to certain decompositions already met in the theory of cyclotomy. They involve factors built from roots of unity, and it is important to relate these factors to the prime ideals of the field Ω_m.

Let p be a rational prime satisfying

$$p \equiv 1 \pmod{m}, \qquad p-1 = mm'. \tag{1}$$

Let g be a primitive root modulo p, and form the periods of m' terms from the p-th roots of unity. If $\eta_0, \eta_1, \dots, \eta_{m-1}$ denote these periods, and if ε is a primitive $(p-1)$-st root of unity with

$$\varepsilon^{m'} = \zeta, \tag{2}$$

then the resolvent

$$(\zeta, \eta) = \eta_0 + \zeta\eta_1 + \zeta^2\eta_2 + \cdots + \zeta^{m-1}\eta_{m-1} \tag{3}$$

is a whole algebraic number. It satisfies a product formula of the form

$$(\zeta, \eta)(\zeta^{-1}, \eta) = \pm p, \tag{4}$$

and hence is a divisor of p.

There are further cyclotomic factors, denoted in Weber's notation by $\psi_s(\zeta)$. They also divide p, and their products with the corresponding conjugates are again equal to p, whenever the exceptional congruences are excluded:

$$\psi_s(\zeta)\psi_s(\zeta^{-1}) = p. \tag{5}$$

The powers of the resolvent are expressible in terms of these factors. In particular,

$$(\zeta, \eta)^m = \pm p\, \psi_1(\zeta)\psi_2(\zeta)\cdots\psi_{m-2}(\zeta). \tag{6}$$

Let $\mathfrak{p}_n$ be the prime ideals over p described in the preceding section. The congruences for ζ modulo $\mathfrak{p}_n$, together with the congruences for the binomial coefficients used in the construction of the ψ_s, determine exactly which $\mathfrak{p}_n$ divides which $\psi_s(\zeta)$. If n' is defined by

$$nn' \equiv 1 \pmod{m}, \tag{7}$$

then a simple counting of residues gives Kummer's factorization

$$(\zeta, \eta)^m \sim \prod_n \mathfrak{p}_{n'}^n, \tag{8}$$

where n runs through the positive integers smaller than m and prime to m. The exponents are taken from the representatives n, and n' is always the inverse of n modulo m.

This is Kummer's theorem in the form needed below. It says that the ideal factorization of the m-th power of the resolvent is completely determined by the inverse residue classes modulo m.

A slightly more general form is useful when $p-1$ is divisible only by a proper divisor of m. Let m_1 be the greatest common divisor of $p-1$ and m, so that

$$m = m_1 m_2.$$

The splitting of p in Ω_m is then the same as in the smaller field Ω_{m_1}, and the prime ideals satisfy

$$\mathfrak{p}_n = \mathfrak{p}_{n+sm_1} \qquad (s = 0, \ldots, m_2 - 1).$$

Applying the preceding formula in Ω_{m_1}, and then lifting back to Ω_m, gives a cyclotomic number Q_n such that

$$Q_n^m \sim \prod_t \mathfrak{p}_{t'}^t, \tag{9}$$

with t running over the reduced residues modulo m, and t' determined by $tt' \equiv 1 \pmod{m}$. The statement applies simultaneously to products of prime factors belonging to rational primes congruent to 1 modulo q, or modulo 4 when $q = 2$.

These formulae are the bridge between the explicit cyclotomic resolvents of the first volume and the ideal factorization in the cyclotomic field.

§175. The Roots of Unity in the Field Ω_m

For the later applications a more exact knowledge of the units of the cyclotomic field is required. We first discuss only the numerical units, that is, the whole numbers of Ω_m whose reciprocals are again whole numbers, or equivalently whose norm is ± 1.

Among the units are certainly the roots of unity contained in Ω_m. We shall show that no others of this kind occur beyond those already forced by ζ. Every root of unity in Ω_m is, up to sign, a power of ζ.

Let ρ be a root of unity contained in Ω_m, and write

$$\rho = \Phi(\zeta) \tag{1}$$

as a rational function of ζ. Let q^a be the highest power of q dividing the order of ρ. Then

$$\rho^{q^a} \tag{2}$$

is a root of unity whose order is prime to q. By the irreducibility of the cyclotomic equation in the extended sense, the substitutions

$$\zeta \mapsto \zeta^n \qquad (n, m) = 1$$

may be applied to the expression in (2) without changing its value. Hence ρ^{q^a} is rational. Being a rational root of unity, it is equal to $+1$ or -1. Therefore ρ, apart from its sign, has order a power of q, and so is a power of the primitive m-th root ζ. Thus the roots of unity in Ω_m are precisely

$$\pm\zeta^k. \tag{3}$$

We shall also need the following theorem of Kronecker. If α is a whole algebraic number and if all conjugates of α have absolute value 1, then α is a root of unity.

Indeed, the absolute value of the norm of a whole number is the product of the absolute values of its conjugates and is a rational integer at least 1, unless the number is zero. Under the present hypothesis the norm is therefore ± 1, so α is a unit. The same is true of every power of α.

It remains to see that only finitely many whole numbers can have all conjugates of absolute value 1. Let

$$\omega_1, \ldots, \omega_n$$

be a basis of the ring of integers, and write

$$\alpha = a_1\omega_1 + \cdots + a_n\omega_n, \qquad \beta = b_1\omega_1 + \cdots + b_n\omega_n.$$

If two such numbers α, β are not equal and not negatives of each other, then among their conjugates there is no real quotient equal to ± 1, and hence

$$|\alpha \pm \beta| < |\alpha| + |\beta| = 2$$

for at least one choice of sign in every conjugate. Thus $(\alpha \pm \beta)/2$ cannot be a whole number, since its norm would have absolute value < 1. Consequently two distinct numbers of this type cannot have all their basis coefficients congruent modulo 2, except for the pair α and $-\alpha$. There are only finitely many residue classes modulo 2, and so only finitely many such whole numbers.

Now the infinite sequence

$$1, \alpha, \alpha^2, \alpha^3, \ldots$$

consists entirely of numbers of this finite set. Two terms must coincide; hence

$$\alpha^{h-k} = 1$$

for some $h > k$, and α is a root of unity.

It follows in particular that if a number of a field is a root of unity of degree m, then all of its conjugates are roots of unity of the same degree. For they satisfy the same rational equation

$$x^m - 1 = 0.$$